Limits & Derivatives

ANSWER KEY

Section 1: Limits & Indeterminate Forms

1 Evaluate the limit: $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

a. 6 ✓

b. 3

c. 0

d. Does not exist

Direct substitution gives $\frac{0}{0}$. Factor numerator: $\frac{(x-3)(x+3)}{x-3} = x + 3$. Substitute $x = 3$: $3 + 3 = 6$.

2 Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{4x^2 + 5}{2x^2 - 3x}$.

a. 2 ✓

b. 4

c. ∞

d. 0

Divide leading terms: $\lim_{x \to \infty} \frac{4x^2}{2x^2} = \frac{4}{2} = 2$.

3 Evaluate the trigonometric limit: $\lim_{x \to 0} \frac{\sin(5x)}{x}$.

a. 5 ✓

b. 1

c. 0

d. $\frac{1}{5}$

Standard limit rule: $\lim_{x \to 0} \frac{\sin(kx)}{x} = k = 5$.

Section 2: Derivative Rules & Tangent Slopes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 If $f(x) = 4x^3 - 5x^2 + 2x - 7$, what is $f'(x)$?

a. $12x^2 - 10x + 2$ ✓

b. $12x^2 - 10x$

c. $4x^2 - 5x + 2$

d. $12x^3 - 10x^2 + 2$

Power rule: $\frac{d}{dx}[4x^3] = 12x^2$, $\frac{d}{dx}[-5x^2] = -10x$, $\frac{d}{dx}[2x] = 2$. Result: $12x^2 - 10x + 2$.

5 What is the slope of the tangent line to the curve $y = x^2 - 3x + 4$ at the point where $x = 2$?

a. 1 ✓

b. 4

c. 2

d. -1

$y' = 2x - 3$. At $x = 2$: $y'(2) = 2(2) - 3 = 4 - 3 = 1$.

6 If $g(x) = (2x + 1)^4$, what is $g'(x)$?

a. $8(2x + 1)^3$ ✓

b. $4(2x + 1)^3$

c. $2(2x + 1)^3$

d. $8(2x + 1)^4$

Chain rule: $g'(x) = 4(2x + 1)^3 \cdot \frac{d}{dx}[2x + 1] = 4(2x + 1)^3 \cdot 2 = 8(2x + 1)^3$.

Section 3: Critical Points & Extrema

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 For what value of x does the function $f(x) = x^2 - 6x + 8$ have a local minimum?

a. $x = 3$ ✓

b. $x = 6$

c. $x = -3$

d. $x = 4$

Set $f'(x) = 0$: $2x - 6 = 0 \Rightarrow 2x = 6 \Rightarrow x = 3$.

8 What is the maximum value of $f(x) = -x^2 + 4x + 1$?

a. 5 ✓

b. 4

c. 2

d. 7

$f'(x) = -2x + 4 = 0 \Rightarrow x = 2$. Maximum value $= f(2) = -(2^2) + 4(2) + 1 = -4 + 8 + 1 = 5$.

9 How many critical numbers does $f(x) = x^3 - 3x$ have on the real line?

a. 2 ✓

b. 1

c. 3

d. 0

$f'(x) = 3x^2 - 3 = 0 \Rightarrow 3(x^2 - 1) = 0 \Rightarrow x = \pm 1$ (2 critical numbers).

Section 4: Higher Order Derivatives & Motion

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 If the position of an object is given by $s(t) = t^3 - 6t^2 + 9t$, what is its acceleration $a(t)$ at $t = 3$?

a. 6 ✓

b. 0

c. 12

d. -6

Velocity $v(t) = s'(t) = 3t^2 - 12t + 9$. Acceleration $a(t) = v'(t) = 6t - 12$. At $t = 3$: $a(3) = 6(3) - 12 = 18 - 12 = 6$.

11 What is the second derivative $f''(x)$ of $f(x) = 2x^4 - 3x^2$?

a. $24x^2 - 6$ ✓

b. $8x^3 - 6x$

c. $24x^2 - 6x$

d. $12x^2 - 6$

$f'(x) = 8x^3 - 6x$. $f''(x) = 24x^2 - 6$.

12 At what x-value does the curve $y = x^3 - 6x^2 + 5$ have an inflection point?

a. $x = 2$ ✓

b. $x = 4$

c. $x = 0$

d. $x = 6$

$y' = 3x^2 - 12x$, $y'' = 6x - 12 = 0 \Rightarrow 6x = 12 \Rightarrow x = 2$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647